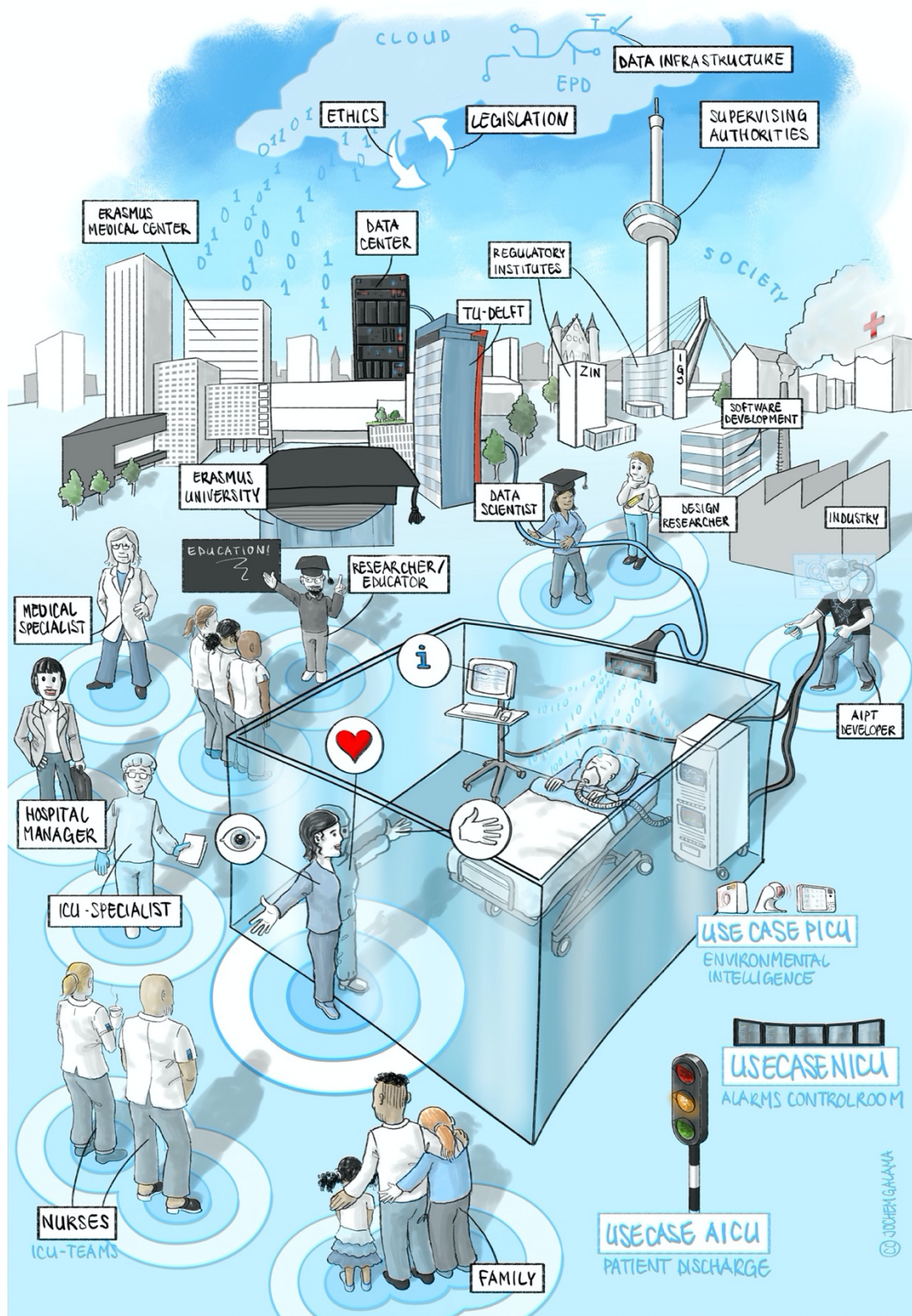


Human-Centered AI for Sustainable Critical Care

Co-designing AI-Powered Technologies in Neonatal, Pediatric and Adult Intensive Care



Human-Centered AI for Sustainable Critical Care: Co-designing AI-Powered Technologies in Neonatal, Pediatric and Adult Intensive Care

1. General Information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

Name*	Institution	Department	Email	Tenured (i.e., fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/Applicant]
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Planned start date of seed funding: 1-9-2022

Primary Convergence theme: *Human Centered Technology and AI for Health*

Secondary Convergence theme (if applicable): *Technology Supported Transitions in Health(care)*

Tertiary Convergence theme (if applicable): *Improving Health Journeys*

Quaternary Convergence theme (if applicable): Choose an item.

Keywords: critical care, nurses, workforce sustainability, human-centered AI, data ethics, participatory design, healthcare governance

1.b Summary (max 300 words)

One of the most pressing problems in contemporary healthcare systems is the growing shortage of healthcare staff, particularly specialized nurses. Beyond nurse education and professional leadership, technological solutions are urgently needed to enhance the sustainability of critical care. In this flagship, which involves neonatal, paediatric and adult ICUs, we shift from mechanically assisted to AI-powered critical care and empower ICU nurses and physicians by giving them adaptive, explainable, and personalized AIPT solutions that are trusted and implementable. In doing so, we address the scientific and empirical innovation challenge of 'bringing AI into the clinic'. Applying a human-centered AI approach, we develop an ecosystem for continuous learning and improvement, targeting both AIPT development (i.e., the inclusion of new data, development and improvement of algorithms and computational models) and its embeddedness in nursing practice. The ecosystem includes a wide variety of hospital actors (critical care teams, patients/family), public regulators (e.g., Inspectorate, healthcare purchasers) and private stakeholders (health-tech industry), and involves a broad range of scientific disciplines

(e.g., medical/nurse science; human-centric AI, data ethics and legislation, AI validation and implementation; science & technology studies; communication science), the focus being to co-design ethical, explainable, human-centered and adaptive AIPT that comply with regulatory and financial requirements. We work with industrial partners and collaborating hospitals to transition from models to clinical critical care practice. Besides AIPT development, we will design a training and educational programme for AI in nursing care to facilitate the shift to AIPT-driven critical care. The ecosystem revolves around three AIPT use-cases that represent current problems in nursing care and enable us to develop and use AIPT in clinical practice: Patient Discharge, Environmental Intelligence, and Alarm Control Room. These use-cases serve as a launchpad for nurse involvement and collective learning, and for further AIPT development to achieve sustainable Critical Care.

1.c Public Summary (max 300 words)

Although the Dutch healthcare system is of excellent quality, the COVID crisis has also revealed its vulnerability, especially in critical care. Medical technology, and more particularly Artificial Intelligence (AI), can support nurses in their work, increase capacity, and enhance quality of care. Research and practice, however, show that nurses (and physicians) scarcely use complex technologies because they do not fit in with their daily routines, sometimes increase rather than reduce their workload, and/or are distrusted. On top of this, use of AI is not (yet) compatible with existing financial arrangements, privacy requirements, and (external) quality regulations. Making AI-Powered Technologies successful requires a comprehensive and participatory approach in which users, developers, healthcare purchasers, and regulatory bodies work together to achieve appropriate and effective healthcare innovation.

In this flagship, we develop a learning and transdisciplinary ecosystem for critical care in which users (i.e., nurses, physicians, patients/family), scientists (e.g., AI and data experts, educational specialists, designers, nurse scientists, science & technology scholars, behavioural scientists), industry, Dutch hospitals and other stakeholders (e.g., the Inspectorate, patient associations) collaborate to create human-centered AI for critical care (i.e., Neonatology, Paediatric and Adult ICUs) and to use and evaluate AI-Powered Technologies in practice. The research revolves around three use-cases that draw on existing research and that capture the current needs of nurses and patients/family: Alarm Control Room, Environmental Intelligence, and Patient Discharge. These AI applications will serve as a launchpad for further AI-development and adoption in critical care practice. We study what financial, ethical, and regulatory frameworks are needed. We evaluate the effects on work satisfaction, quality of care, patient satisfaction, and workload during the research and use our findings to make further improvements. Students at EUR/EMC/TUD are involved in all stages of the research to encourage transdisciplinary knowledge production and innovation.

2. Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)

This flagship addresses one of the most pressing problems in contemporary and future healthcare systems, both in the Netherlands and internationally: **the shortage of critical healthcare staff, particularly specialised nurses**. The increased demands on healthcare delivery requires autonomous capacity growth (staff, training, facilities, and finances), challenging the quadruple aim of better health outcomes, improved patient experience, improved staff experience, and lower cost of care[1]. The COVID pandemic has stirred an immediate call to action and shown the importance of flexibility in transforming healthcare processes to meet heavier and new care demands[2]. The WHO argues that technological solutions can promote a sustainable healthcare sector[3]. Artificial Intelligence (AI), which offers novel tools to augment socio-technological workspaces and care delivery, can meet these demands.

AI Powered Technologies (AIPT) (e.g., adaptive/predictive/explainable/personalised information that can be used in patient care, organisation of care processes, and/or work environments) are a promising method of enhancing healthcare sustainability, especially in technology- and data-driven contexts such as critical care [4, 5]. Clinical uptake is limited, however, due to regulatory, technological, and human constraints[6, 7]. AI-powered innovations are hampered by privacy, quality, and financial regulations as well as inadequate data infrastructures and lack of performance (generalizability, explainability) and validation in a real clinical setting, and hence lack of trust in AIPT among users, both healthcare professionals and patients[8]. This flagship takes **a comprehensive and innovative approach by bringing together healthcare professionals, patients/family, regulators, healthcare purchasers, industry and scientists to co-design explainable, human-centered and adaptive AIPT for critical care**. We co-design, test, and clinically validate AIPT to (1) meet professionals' and patients' needs; (2) improve the accessibility and quality of care for patients and family; (3) augment clinical and nursing capacity; (4) optimise the work and healing environment at the Neonatal Intensive Care Unit (NICU), Paediatric Intensive Care Unit (PICU) and adult ICU (AICU) (referred to jointly as ICUs).

Promises and Pitfalls of AIPT

Despite the great promise of AI, few AIPT have taken it to the bedside[3]. In fact, 90-94% of AI-models in critical care get stuck in the development and prototyping phase, with retrospective models being expanded instead of progressing to clinical implementation[9]. Barriers to deployment have shifted from technological hurdles (i.e., poor fit between users and technology) to regulatory obstacles and ethical and legitimacy issues around emerging data infrastructures[7, 10]. In addition, few healthcare practitioners (especially nurses) or patients have been involved in AIPT development[11-13], with a few exceptions[14]. The expectations and fears induced by their lack of involvement play a critical role in AIPT adoption and creates uncertainty in AI-driven product development for healthcare[15, 16].

Ecosystem for Participatory Design & Adoption

Bringing AIPT into the clinic requires a transdisciplinary and Community approach ensuring that AIPT meets professional/patient/family needs and is accepted and supported within the broader healthcare network, including (quality) regulation and purchase of care [3, 18]. In the current systems, patients (and families) do not know what to expect from AI, and regulators have great difficulty governing AI use[19, 20]. We apply **a participatory and human-centered approach to AI**[11, 21] in which we co-design AIPT with the end-users –nurses (also as co-applicants of this proposal), physicians, allied healthcare professionals, patients, family— and public/private

stakeholders (Figure-1). The resulting **ecosystem for participatory design and adoption** unites these users and stakeholders, each with its own expectations, resources, and interests, in co-designing AIPT that are feasible, adaptable, and explainable and shapes a context for mutual learning, adjustment and adoption. We share lessons learned regarding content and process with partner hospitals (as part of this flagship and as a dissemination activity) and scientists (through publications, conferences, seminars, and protocols). These activities extend our local community approach to the broader healthcare system, encouraging the development and use of AI in healthcare.

Our ambition is to modernise critical care through a digital transformation in which AI becomes accessible, explainable, acceptable, usable, and deployable. Our approach will also **facilitate improvement of AIPT**, by providing an environment for the continuous improvement of AI-models by including new data and stakeholder feedback. AIPT will not replace human care; care provision is both technical and social and relies on contact between people. Tacit knowledge—embodied knowledge that is hard (or even impossible) to define—is an important part of healthcare work and based on embodied experience and on-the-job learning[22, 23]. How AIPT interacts with embodied knowledge and reconfigures care routines grounded in compassionate ‘human’ interaction is a focus of this research.

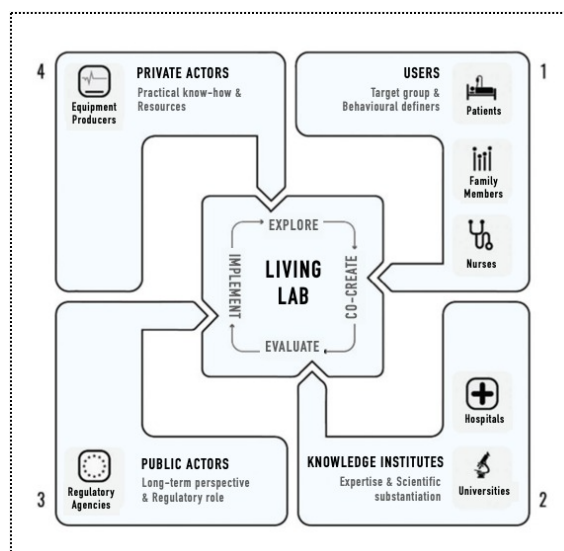


Figure-1: Overview Living Lab Community[17].

Contribution to H&T Mission

In this flagship, scientific innovation and healthcare improvement go hand-in-hand in targeting the convergence aim of enlarging **societal impact**. We are **making critical care more accessible** (a pressing problem in the Netherlands during the COVID pandemic) by improving the work environment for nurses and creating efficient and high-quality care routines and processes when attending severely ill patients. Improving nurse availability and physician-nurse collaboration sustains workforce capacity and thus contributes to **socio-economic equality in healthcare** (‘care available to all in need’).

An important mission of H&T Convergence is **scientific innovation**. We are developing a participatory and inclusive approach to AIPT development and evaluation involving a wide range of hospital actors and stakeholders not normally involved in AIPT development: nurses/patients/families, ethicists, purchasers, and regulators. Participatory methods of co-design, ethical consultation, and deliberation encourage mutual learning and result in feasible, trustable, and acceptable AIPT. This will result in an ecosystem that facilitates continuous

learning, and in which AIPT are constantly being improved based on the needs and input from clinical practice. Our innovations also extend to **education**, as we engage **bachelor and master students at every stage** (i.e., data training, stakeholder mapping, co-facilitating reflexive sessions and workshops; see Workplan). We work with students on developing new approaches to virtual/mixed/augmented reality in nursing education and training, leading to more efficient and attractive nurse/physician training methods with spill-over effects in other domains. Another key aim is to **bring AIPT into healthcare practice**, first in the living lab and, under GDPR approval, in the ICUs. We involve hospitals and scientists in and outside the Netherlands to promote **upscaling and dissemination**.

Finally, we are uniting talented researchers at every level, from bachelor students to top scientists, thereby creating a highly ambitious yet accessible and inclusive environment for academic careers aimed at **transdisciplinary 'convergence' academic positions** at assistant/associate/full professor levels.

2.b Synergy between disciplines

The ErasmusMC (EMC) ICU is an acknowledged frontrunner in AIPT development, with the Care Technology Group and ICU-Data Hub focusing on human-sensitive use of data-driven technologies and AI to support clinical decision-making and management of clinical processes[24-27]. The innovative computational data infrastructures, expertise and experience allow us to take the next step in AIPT development and clinical use. **TUD's Human-centered AI Group** establishes the groundwork for developing transformative AIPT behaviour in the clinical context (adaptive-AI), using historical data about clinical conditions and outcomes to extrapolate the future (predictive-AI), tailoring AIPT to the preferences and needs of users (personalised-AI) and creating mechanisms guiding AIPT behaviour that explain patterns and cause-effect relationships in clinical workflow environments to users (explainable-AI)[4][28][29]. The Erasmus School of Social Behavior (ESSB), the Erasmus School of Health Policy & Management (ESHPM) and the Erasmus School of Law (ESL) at **Erasmus University Rotterdam (EUR)** provide the social sciences perspective by assembling experts in communication, ethics, law, science & technology studies, healthcare governance, logistics and change management, delivering the expertise required for an in-depth understanding of AI adoption in highly complex organizational and policy settings[30]. Some EUR scholars are experts in action-research[31][32], an underlying mechanism for our participatory and community approach.

2.c Long term sustainability plan

The many challenges involved in 'bringing AIPT to the clinic' within critical care call for the ongoing involvement of hospital actors, data scientists and managers, and public and private stakeholders. Addressing differing needs and taking time to solve emerging (unforeseen) difficulties require commitment, endurance, and close involvement of the community, as well as stable financial resources and a thriving academic infrastructure (ongoing funding, career opportunities, education). AIPT development and implementation also take longer, as research and practice consistently reveal the difficulty of reaching higher TRL levels, gaining trust and acceptance among users, and consolidating data infrastructures[3].

Risks and solutions:

- *Stakeholders may wish to opt out due to conflicting interests:* Regulators need practice-based experience to develop and adjust their regulatory arrangements for AI. Representatives from the Inspectorate (IGJ) and the Dutch Healthcare Institute (ZiN) have formal academic positions at EUR; the private companies involved similarly aim to learn from practice-based experience. Their involvement is regulated through formal agreements, including ethical criteria.
- *Ethical and legal regulations may hamper the transfer of AI-models and prototypes to the clinic:* An important aspect of our flagship is the Data and Model Management Platform, consisting of data and ethical and legal experts at the EMC/TUD/EUR. They guide the development process in line with formal requirements.
- *Resurgence of COVID-19:* The pandemic has exposed and aggravated the shortage of IC nurses; if a new outbreak occurs, we will continue our research while taking safety precautions, i.e., video-conferencing.
- *Lack of funding:* AI is key in policy and research funding agendas (see, e.g., the NLAIC in the Netherlands, the NWO AI research agenda, and the Horizon Europe programme) for its implementation in society. To date, all three institutions have been very successful in obtaining research grants and are widely acknowledged for their expertise. We believe we can build on this strong basis and obtain more prestigious grants for both individual scholars and our transdisciplinary group.

2.d Visibility, attracting talents and new partners

This flagship's thriving living lab community brings together ErasmusMC's ICUs, industry, scholars, government bodies, and other hospitals. One early example is IC 2.0, a current EMC initiative fostering data-centric care, including the new IC datahub. The community is a magnet for talent and other Dutch and international partners. Relationships with Treant and Maastricht UMC support our negotiations with other hospitals. All EMC/TUD/EUR departments involved have outstanding reputations in their fields, reflected by prestigious grants (ERC and Horizon 2020) and publications in top journals. TUD scholars involved in this flagship co-manage the AI programme Data, AI, and Digitalisation and national ICAI network, both examples of high-performance and scientific excellence in AIPT. Another example is the EUR's AiPact, focusing on transdisciplinary human-centered AI. Its embeddedness in the Erasmus Center for Data Analytics (ECDA), including all EUR faculties, ensures strong ties with private partners. Finally, there is AiPact/ECDA's role in EUR's Opening of the Academic Year 2022/23, which will showcase this flagship research.

All flagship consortium partners have well-established relationships with scholars around the world and private and public partners. The ICUs work closely with ITeMedical, Dräger Patient Monitoring Systems, SAS, Chipsoft, MedPac, HealthRI, LeQuest and Philips Healthcare, allowing us to attract co-funding and exploit their expertise in co-developing AIPT. Successful collaboration (staged in publications, seminars, and conference presentations) will attract more partners.

Besides private partners, the consortium has well-established relationships with government and intermediaries. ESHPM collaborates with the Inspectorate for Healthcare & Youth (*Inspectie Gezondheid&Jeugd, IGJ*) and the Dutch Healthcare Institute (*Zorginstituut Nederland, ZiN*), which both play a central role in AI regulation and funding. We will use these networks to involve more public and private actors (formalized in WP1 of this flagship).

All consortium partners have attracted talented national and international researchers at all academic levels, including bachelor and master students. For example, TUD's Industrial Design Engineering is renowned for its Medisign Education Programme for MSc and PhD students, providing a solid background for collaboration with hospitals and health-tech companies. The EIT BeNeFit programme for innovation fellows is a partnership between the three universities and KULeuven. The flagship has ample room for students at all levels and offers internships ('Student Dream Team', see below) in which they help coordinate the Living Lab. Students will have the opportunity to obtain interdisciplinary training and research experience in their graduation research (Ba/BSc/Ma/MSc theses, all consortium partners); and we develop a transdisciplinary summer school for PhD students, building new networks for interdisciplinary talent development. We encourage transdisciplinary academic careers through the summer school and related activities (i.e., actively supporting junior-researchers in grant applications). See Table-1 for our planned funding activities.

Table-1. Active funding possibilities.

Active funding possibilities Estimated Total (Euros)			
NWO	EU	Partner Institutions	Industry
Data & Ethics Infrastructures for Human-centered AI for Health VICI / NWO to be submitted in August 2022 250K can be used for Flagship	Development pathways for public-private partnerships Horizon-Widera-ERA To be submitted in April 2022. 175K can be used for Flagship	TU Delft Education Fellowship To be submitted in 2023 50K (for 2 years)	Edge Federated AI (Cognizant) to be consolidated for 200K
Environmental Intelligence by Sound Open Technology Program / NWO to be submitted in June 2022 250K can be used for Flagship	Smile4Safety- An AI-Coach for enhancing the quality of care and patient safety HORIZON-HLTH-2021-CARE-05-01 submitted in 21-09-2021 175K can be used for Flagship	EUR Communication for Learning in Innovation (CLI) To be submitted in 2023 50K (for 2 years)	Co-design processes for AI-powered Applications in Critical Care (Philips Design) - to be consolidated for one PhD project in 2025 250K
NWO-ORC 2022 AI4Access Human-Centred AI techniques to facilitate access to knowledge and services pre-proposal to be submitted in June 2022 150K can be used for Flagship	Speculative futures of AI-Powered Nursing to be submitted in 2023 ERC Consolidator Grant 525K can be used for Flagship		
	AI-Supported Interactive Hybrid Decision-Support Systems for Fair Recruitment in Tertiary Education HORIZON-CL4-2022 -HUMAN-01-01 to be submitted November 2022: AI for human empowerment (AI, Data and Robotics Partnership) (RIA). 200K can be used for Flagship		
650K	1,075 million	100K	450K

2.e Long term societal impact

Moonshot (Impact)

Our aim is sustainable (i.e., affordable, high-quality and well-staffed) Critical Care through a **paradigm shift from mechanically assisted to AI-powered Critical Care** through **The Internet of Things**, with nurses and physicians feeling supported and in control of the delivery of high-quality and safe care to critical patients, and with patients and family experiencing a low-stress and highly competent medical and nursing environment.

Outcomes (> 5 years)

What

- Augmented decision-making through AI-supported predictive modelling, precision medicine and environmental control for nurses and physicians in critical care.
- Low-stress critical care environment for healthcare professionals, patients, and family.
- State-of-the-art, individualised education and training programmes for AI in critical care targeting nurses, physicians, and allied healthcare professionals.
- Adequate hospital-based and public regulatory, financial and cost-effectiveness systems that control and encourage ethical AIPT development and use, supported by patient, professional, and hospital associations as well as quality assessment agencies.
- Transdisciplinary (convergence) academic master programme on human-centered AI in healthcare.

How

- Thriving and well-supported transdisciplinary academic community for human-centered AI-development in healthcare that is successful in obtaining prestigious individual and collective research grants and offers attractive career pathways for transdisciplinary research in AI.
- Learning community of collaborating hospitals in the Netherlands and beyond on AIPT development and use creating an ecosystem for continuous learning, and the responsible implementation of AI in clinical practice.
- Well-resourced collaboration with industrial parties that fosters continuous development and learning, with firm strategy and agreements about ethical procedures and property rights.
- Institutionalised relationships with public regulators and healthcare purchasers for AIPT development and use.

Output (5 years)

- Guideline for ethics and regulatory approval (e.g., CE mark).
- Sensor systems and infrastructure for computing power and smooth data flow (patient-centric and environmental sensors).
- Adaptive dashboards with summarised data for public/general use.
- Educational/training modules and competence framework for AIPT use in critical nursing care settings.
- Evaluation and implementation framework for human-centered AI-development in critical care, engaging healthcare professionals and patient/family.
- Master Classes on human-centered AI for healthcare professionals and managers.

Desired TRLs of AIPT

Our past and current AIPT efforts have already produced novel concepts (i.e., AI coach for supporting clinical decision-making), prototypes (i.e., soundscape dashboard; alarm distribution and visualisation), AI and signal processing algorithms (i.e., DALC for deep active learning from humans, SECA for AI explanation by human concepts and rationales, Marta for integrating human rationales into AI, Scalpel-HS for characterising AI unknowns, OpenCrowd for training data creation and Scalpel-CD for data cleaning, accurate auditory detection of alarm signals), and

further products from our industrial partners (i.e., alarm management and distribution). Prior TRLs range between TRL2 and TRL5. Our prior work gives us a head start but we also aim to develop novel concepts resulting in product/service/system design prototypes (TRL1-4). It will be feasible (in WP3 and WP6) to move up to TRL7 (system validated in a clinical environment) with contributions by industrial and hospital partners. The ambition is to reach TRL8-9 with the support of the CE-marked industrial partners.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

Our **flagship ecosystem** (see frontpage) has the momentum to make critical care human-centered, data-driven, safe, and sustainable. Nurses and patients/families (e.g., NICU/PICU parents) are the primary users and beneficiaries of healthcare technologies in ICUs. Our participatory and inclusive approach is at the forefront of Health & Technology research as widely advocated[3] but—to our knowledge—not yet applied.

Our flagship applies AIPT to transform data into (adaptive/predictive/explainable/personalised) information to use in patient care, organising care processes, and work environments. AIPT include machine-and deep-learning but also network models that monitor and analyse clinical processes with shared mental models, supporting clinical decision-making by predicting and modelling precision medicine. In the next decade, integration with robotics, virtual and augmented reality, and machine-to-machine communication[33] is expected to change healthcare irrevocably. **AI-Powered Technologies will transform and improve critical care** by:

1. Improving the quality of **clinical definitions** (i.e., medical precision, personalised medicine) and care processes (i.e., saving bed-days with prediction models for discharge, lowering work pressure with robotic process automation (RPA) for administrative tasks)[4][27];
2. Optimising the work and healing environment by **explaining** cause and effect (i.e., fewer alarms facilitate patient sleep)[13][34];
3. Improving nurses' work efficiency and satisfaction by **augmenting** nurse cognitive capacity (i.e., seeing patterns in patient events in terms of alarm actionability)[35].

We develop our ecosystem and establish its potential in **three use-cases**: *Patient Discharge*, *Environmental Intelligence* and *Alarm Control Room* (Table-2). They will teach us about multi-actor/stakeholder **co-design and AIPT development and use in critical care** and function as a launchpad for further AIPT-development in critical care, establishing a foundation for the flagship in the longer term. We will develop and evaluate the AIPT iteratively in increasing complexity, starting from passive visual displays to more complex AI-machines-and-networks that allow for predictive/personalised medicine, ultimately leading to adaptive dashboards, awareness tools and smart assistants for medical/nursing decision-making.

Table-2. Use cases and their contribution to the AIPT development per ICU.

USE CASES			
Relevance	Patient discharge	Environmental Intelligence	Alarm Control Room
Clinical Context	Adult ICU	Paediatric ICU	Neonatal ICU
Aim	Predict whether it is safe to discharge a patient (<i>predictive medicine, personalised medicine</i>)	Monitor, define and predict environmental stressors (e.g. air quality, temperature, sound types)	Significantly reduce the number of false, non-actional medical and technical alarms
Contribution to workflows (Nurse / Medical) and Wellbeing (Nurse Patient / Family)	Improved team dynamics through well-informed decision making	Improved patient – family well being / comfort	Improved alarm-induced nurse actions and reduced stress and workload
Specific technical contribution to Human-centered AI development	DEFINE Develop robust ML algorithms that meet the social / ethical / legal / privacy etc. constraints	EXPLAIN Make the critical care environment explainable, through responsive environmental awareness by AI-powered tools	AUGMENT Augmenting nursing work by AI-powered tools for better care provision with smart assistants
Co-design	Focus on multidisciplinary teams in AIPT development and evaluation	Focus on engaging patient & family in AIPT development and evaluation	Focus on nurses in AIPT development and evaluation
Knowledge transferable to other ICUs/ partner hospitals	In-depth insight in the relation between the network of distributed knowledge and the use of AIPT	Insights into cause-effect relationships and clinical patterns that cater for improved patient wellbeing	Insights into perceptual and cognitive capacity of nurses and human-human and human-machine interactions

3.b Aims and feasibility

This flagship establishes a thriving community in which participatory design of AIPT boosts the **sustainability of critical care by improving nurse capacity, working conditions and patient safety and deepens our knowledge of how to design-develop-deploy-use and improve human-centered AI in critical care.**

Specific aims and funding are identified in the Work Packages below, including a plan for student inclusion (WP1&4). WP2&3 attract industrial co-funding (in-kind/cash); WP3&5 will attract additional research funding, WP4 attracts educational grants, industrial co-funding; WP6 attracts hospital co-funding and feeds into WP1 for community relations. Ideally, we host ten PhD and Postdoc projects. We will also work with existing PhDs/Postdocs without co-financing requirements, giving us additional staffing. Student assistants will support data-mining/modelling/prototyping and community management.

3.c Work plan

The main question guiding the flagship research programme is: **How can human-centered AIPT for critical care be adopted (i.e., ethically & legally co-created/tested/used/evaluated) in nursing/medical practice in close collaboration with healthcare professionals, patients/family, and public/private stakeholders in order to enhance the sustainability of critical care?**

The work is carried out in seven work-packages (WP), each with a methodological approach complementing our aim to establish AIPT for critical care (see Figures-2&3). Continuous design and evaluation reveal the change process spurred by the paradigm shift towards AI-Powered critical care and the impact on healthcare professionals, patients/family, and outcomes. Our WPs promote co-design with the critical care community(WP1), safe (legal/ethical) infrastructure for data management and modelling(WP2), AIPT development and testing(WP3), incorporation of AIPT knowledge and practice into professional education and training(WP4), continuous evaluation of AIPT users' adoption and acceptance and how this influences existing work practices, nursing routines and patient experience(WP5), dissemination of knowledge and AIPT applications to the critical care community(WP6), and project management(WP7).

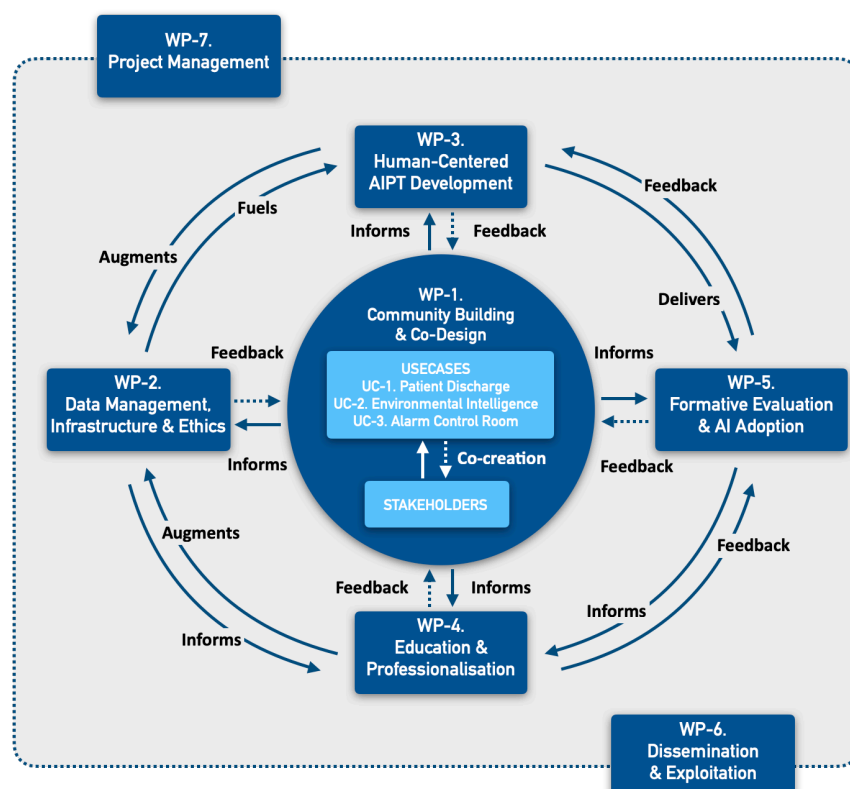


Figure-2. Work-packages work as a platform for continuous learning via co-design/development/evaluation/deployment of AIPT-applications.

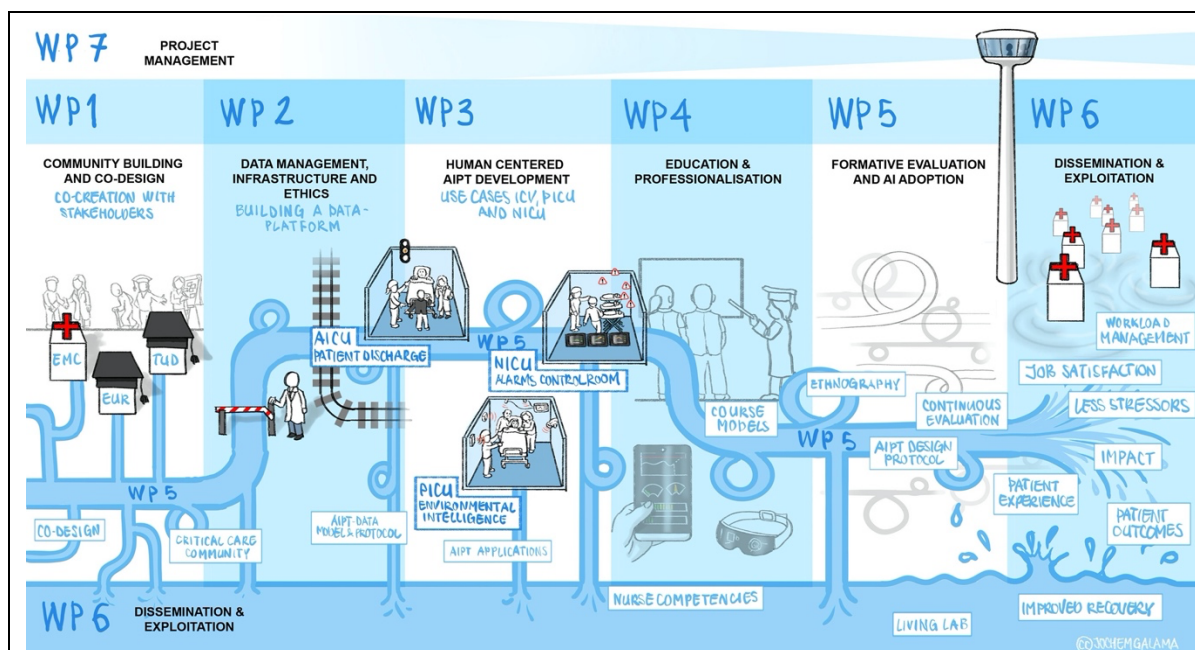


Figure-3. Flagship programme workflow.

Our flagship uses sensitive patient data in collaboration with industry, thus, we adhere to GDPR, open science, and hospital data privacy rules by considering ethical/legal/storage aspects(WP2). Regarding local use of patient and hospital data for AIPT development, private partners are chosen on the merits of their CE-marks for AI-development and data management. Our research adheres the FAIR principles (Findable/Accessible/Interoperable/Re-usable data) for the community, with possible restrictions on public sharing of data.

WP1: Community Building & Co-design

The **objectives** are to establish the Living Lab (LL), offering a dynamic, inclusive, participatory learning environment for all hospital actors (nurses/physicians/allied professionals/patients/family) and public/private stakeholders (*I/GJ*-Inspectorate for Healthcare&Youth; *ZiN*-Dutch Healthcare Institute; patient/family representatives, health insurers, industrial partners). WP1 is also a meeting/communication platform.

Approach: WP1 revolves around the three use-cases providing input for AIPT-development and organises reflexive sessions with public/private stakeholders to facilitate learning and understanding. The LL is a design platform and testbed for AIPT encompassing public regulation, private finance, patient/family/professional needs, and industrial requirements. It is coordinated by senior researchers and students in the multidisciplinary Student DreamTeam (internship model), annually attracting TUD-EMC-EUR students.

Work Package 1		
Main Lead EUR	Partners EMC & TUD	Interconnected Work packages WP3 – WP5 – WP6
Main Aim	To facilitate the co-design (development, testing and use) of AIPT for critical care, investigate how public (i.e., patients, regulatory agencies, healthcare professionals) and private (i.e., industry) collaboration is maintained and continuous learning is organised and sustained	
Aims		Output
- Stakeholder mapping public and private stakeholders (interviews, document study) - Living Lab: organise and maintain physical and virtual meeting space and communication platform - Select, manage Student Dream Team - Organise, analyse reflexive sessions for development and learning (twice yearly) - Define design requirements for the selected use cases in the Living Lab - Create protocols for multi-stakeholder collaboration (including IP ownership, ethics of dataset usage and appropriation)		- Design requirements for use cases - Insights and protocols for professional and patient education and professional/patients/family engagement in human-centered AI creation (with WP5) - Protocols for participatory design processes in healthcare (with WP5) - PhD thesis, Scientific publications, conference attendance

WP2: Data Management, Infrastructure & Ethics

The **objective** is to establish a **Data and Model Management Platform** (DMMP) for creating datasets and safe (ethical/legal) AI-tools. DMMP ensures compliance with norms/standards of data and AI ethics development and use and broader constraints, e.g., the patients' and professionals' rights, stemming from the law and professional ethics. Critical care is also a professional environment for nurses/physicians, so employment ethics and legislation influence data-collection and -processing practices and deployment of AI-driven systems. In collaboration with industrial partners, WP2 implements principles/rules for data model lifecycle management (i.e., reconfiguring the EMC-SAS data management platform), as well as a software platform for data management and visualisation.

Approach: We implement state of the art approaches to *i)* lifecycle management of data and AI-models for healthcare; *ii)* legal/ethical aspects of data collection; *iii)* trustworthy AI-model development and cost-sharing schemes for AI-models in healthcare. We will establish a **Data and Model Management Board** (DMMB) to oversee the DMMP, with experts in IT-law and ethics, IP-law, data protection, AI, and other contexts (Digital Humanities, Philosophy, Political Science, Sociology). External data protection and research ethics specialists will join representatives of healthcare professionals and patients/family on the DMMB, which will deliver internal services to address ethical and legal problems in collecting/storing/sharing/utilising patient, medical and logistics data.

Work Package 2		
Main Lead TUD	Partners EUR & EMC	Interconnected Work packages WP3 – WP5 – WP6
Main Aim	Define the legal and ethical requirements for data collection, creation and use for AIPT production and management, and constraints for data organisation that researchers and healthcare providers act in accordance with this diverse set of laws and regulations that are continuously scrutinised and revised at hospital, national and international levels.	
Tasks		Output
- Develop ethics and legal framework for data collection and AI model design for healthcare applications (and ICU in particular), with guidance from the DMMB. Design guidelines on how to choose cost functions and create real and synthetic datasets for training AI models, in addition to evaluation metrics (e.g., bias) and procedures, taking into account ethical and legal considerations - Configure the data, model, and applications funnels in the SAP & Microsoft Azure enabled Datahub at EMC following the developed ethics and legal framework. This Datahub platform is connected to numerous sensor systems and other data sources across the EMC - Computational resources for AI model development and execution. - Design of dashboard as interface for the SAP Datahub, where data is summarised and synthesised for general use - Prepare the necessary documents and guidelines for ethics applications - Investigation of potential cost sharing schemes for AI-based services in the specific context of ICUs		- DMMP and DMMB - Guidance for ethics approvals (applications and reporting); regulatory approvals (e.g., CE mark) - Liaison with EMTC committees (ethics, SAP/data management, etc.) - Cost sharing schemes - Sensor systems and infrastructure for data flow (patient-centric and environmental sensors) - Hub for data collection, computing power and displays for a smooth data flow - Dashboard with summarised data for public/general use - PhD thesis, Scientific publications, conference attendance

WP3: Human-Centered AIPT Development

WP3 spans the three ICUs. Within each, a specific use-case is co-designed with the representatives of primary/secondary users (See Table-1). The **objective** is to respond to the needs of primary/secondary users (nurses, teams, patients/families). We design interpretable AIPT for efficient human-machine and human-human reliance and implement machine-learning models, networks, and explainable AI techniques to enhance AIPT users' perceptual/cognitive expertise and their experience of the environment and workflows.

Approach: We follow *AI-Modelling cycle processes* to *i)* create value-sensitive and (combined) metrics for assessing AI systems and networks in critical care; and *ii)* measure accuracy of definitions, explanations, and predictions (see Table-1). We follow the *continuous software engineering cycle processes* implementing AI models in the ICUs to *i)* plan-code-evaluate-build-test AIPT-systems and *ii)* release-deploy-operate-monitor AIPT in the ICUs. Finally, we will evaluate the usability of and ease of interaction with simple and more complex AIPT solutions. In terms of AIPT systems we design deep contextual neural signal-vision-language models (e.g., multi-modal Transformer) adaptable to IC-tasks based on large clinical data (e.g., longitudinal HER data, vital-signs, patient history/records, environmental parameters (sound-light-temperature levels/categories), alarm-frequency).

Work Package 3		
Main Lead TUD	Partners EMC & EUR	Interconnected Work packages WP1 – WP2 – WP4 – WP5 – WP6
Main Aim	Align needs for AIPT-development and use to reduce perceptual and cognitive load as well as interrupted workflows while increasing the quality, effectiveness, and outcomes of care, as well as trust in AIPT among healthcare professionals and patients/family	
Tasks		Output
- Identify challenges related to the urgent needs in workload and stress reduction while improving care quality. - Define an AI development and maintenance roadmap with the stakeholders to ensure the regulated and trustworthy co-design and evolution of AIPT systems. - Develop machine learning models for filtering and distributing actionable alarms, and for forecasting stressors and advising patient sleep. - Develop machine learning interpretability methods that explain machine learning decisions in human-understandable languages and infer model unknowns for accountable decisions and appropriate human reliance. - Analyse the perceptual and cognitive factors that affect interpretation usability and reliance, their co-variation with the experience and expertise of users, and design interpretation language (visual and textual) personalised for users. - Develop and organise structured frameworks for the assessment of AIPT against the dimensions of Fairness, Accountability, Contestability, and Transparency (FAcCT).		- Algorithms, AI machines, data visualizers, adaptive dashboards for the use cases (with WP2) - Treatment-related key performance indicators can directly be measured and personalised suggestions/models can be given at bedside (with WP2) - AIPT with appropriate sensor systems that support real time representation of complex and aggregated information about the context and patients' situation and experience - Personalised and adaptive interfaces and workflows for situational and environmental awareness (with WP4 & WP5) - Explainable AI methods for assisting healthcare professionals in the decision making at the ICU - Human-centered AI co-evolution methods that integrate stakeholders' feedback into the AI systems to create a minimum viable version of the human-AI systems for the ICU (with WP1 and WP5) - PhD thesis, Scientific publications, conference attendance

WP4: Education & Professionalisation

The **objective** is to support the uptake of AIPT in critical care and develop an education/training module and competency framework for ICU nurses/students on (ethical) AIPT utilisation, based on the adjusted NASSS[8] or Normalization Process Theory[36] and other technology adoption models. WP4 will use available extensive (medical) data to promote sustainable learning and expertise development and integrate AIPT into early simulation-based training for nurses using AR/VR/MR tools. Based on these experiences, we offer online MasterClasses to teaching hospitals/nurse training programmes. WP4 reaches out to students in other disciplines by developing interdisciplinary (postgraduate) events, e.g., summer school for human-centered AI in healthcare (with Erasmus Graduate School, ErasmusX project, TUD-Medisign Programme, BeNeFit innovation fellowship). We will also provide a project hub for bachelor/master/PhD students from EUR/TUD/EMC.

Approach: We conduct a *systematic literature review* on AIPT & education in (nursing) healthcare and *qualitative studies* (e.g., ethnographic) to research nurse/healthcare professional-AIPT interaction (with WP2&3&5) and run *task analysis* with experts on AI, healthcare, and education to compose and test a competency framework for AIPT in critical nursing care using VR/AR/MR tools, including *observations and follow-up interviews*.

Work Package 4		
Main Lead EMC	Partners TUD & EUR	Interconnected Work packages WP3 – WP5 – WP6
Main Aim	Define competencies, skills and innovative educational methods that are required to prepare current and future generations of healthcare professionals and students in healthcare to enable human-centered AI development and use in critical care, e.g. through the use of VR/AR/MR methods for learning skills to operate AIPT applications.	
Tasks		Output
- Research nurses' future needs with AIPT (with WP3) - Based on the findings from previous steps, we develop and evaluate training modules for using AIPT tools in critical nursing care. - Develop an interdisciplinary PhD summer school to bring talents from relevant domains together. - Set up a hub for students for AIPT projects. - Develop and use new VR/AR/MR methods for learning.		- Competence framework/s for AIPT use in critical care settings (with WP3) - Educational/training modules for ICU healthcare professionals as nurses working with AIPT - Evidence based recommendations on introduction of AIPT in early simulation based training - Summer School for AIPT in critical care - Hub for TUD/EMC/EUR bachelor and master students theses on AI & healthcare, with combined supervision - Scientific publications, conference attendance

WP5: Formative evaluation & AIPT Adoption

Objectives are to study the processes and (un)intended implications of AIPT development and adoption for sustainable critical care. We evaluate the impact of AIPT on two relevant key indicators: *care outcomes* (i.e., quality of care, caregiver-patient/family interactions) and *work outcomes* (i.e., work load/flow/satisfaction). As the evaluation is formative and conducted during AIPT development, it provides continuous feedback and supports intermediate adjustment/improvement. We will learn how AIPT blends into critical care practice and define a protocol for professional/patient/family involvement in human-centered AI-development.

Approach: We conduct *ethnographic research* into *i)* nursing critical care (NICU/PICU/AICU) to examine nurses' use, needs, and expectations of technology and AIPT and align with current care processes, aims and required competences, and *ii)* the AIPT co-design process in the LL, with participatory observations and semi-structured interviews with nurses/physicians/patients/family to foster user involvement. We will *evaluate* the effects of the three use-cases on nurse/physician–patient/family interaction, teamwork, quality of care, workload (especially perceptual and cognitive load), and patient/job satisfaction using Ethical-constructive Technology Assessment[37] and Experienced Sampling Methods[38]. Semi-structured interviews with nurses, (former) ICU patients and family about NICU/PICU/AICU sensory environments will focus on the perceptual/cognitive load at the ICUs and possible AIPT-driven changes. Finally, *workshops and focus groups* with physicians, nurses, patients/family, and quality managers will enable to develop and use/test performance metrics on patient outcomes in relation to AIPT.

Work Package 5		
Main Lead EUR	Partners TUD & EMC	Interconnected Work packages WP1 – WP3 – WP4 – WP6
Main Aim	Evaluate the impact of co-designed AIPT on critical care practice, the work of healthcare professionals and the experience of patients/family, as well as the conditions for valuable AI-adoption in critical care.	
Tasks		Output
- Three ethnographic case studies (NICU/PICU/AICU) about technological/AIPT needs and expectations - Ethnographic study of AIPT co-design - Evaluation of the three use-cases through Ethical Constructive Technology Assessment and Experienced Sampling Methods - Semi-structured interviews nurses, patients, family about sensory environment - Workshops for developing patient outcome metrics for AIPT use		- In-depth insight in patient/family/nurse/physician needs and experiences, and how AIPT gets blended in existing nursing/medical practice - Protocol for professional/patient/family involvement in human-centered AI development in critical care (with WP1) - Metrics for patient outcomes in relation to AIPT-use in critical care - Based on the former outputs; vignettes for the reflexive sessions (WP1), input for AIPT development (WP3) and education(WP4) - PhD thesis, Scientific publications, conference attendance

Work Package 6 - Dissemination & Exploitation

The **objective** is to involve other national/international hospitals to adopt AIPT in critical care. By creating an AIPT learning community for critical care practice, we disseminate our findings in a “one-to-three-to-all” system, extending use-case solutions to the three ICUs and then to all ICUs in participating hospitals. Other dissemination channels are the ECDA network, scientific/professional conferences, journals/papers. We also establish quarterly MasterClasses to disseminate WP1-5 solutions to professionals (i.e., AI-engineers, healthcare designers/professionals/managers).

Approach. We operate through a network of regional and research hospitals and Holland Health-Tech creative industries to disseminate AIPT, facilitate collaboration between PhD and Postdoc researchers, and scan valorisation opportunities in NL/EU/US.

Work Package 6		
Main Lead EMC	Partners EUR & TUD	Interconnected Work packages All
Main Aim	Disseminate AIPT application and knowledge for critical care to other hospitals and healthcare professionals, in accordance with ethical, professional and other legal requirements, to enable wider learning and development, and sustain future critical care provision.	
Tasks		Output
- Establish an extended living lab including (inter)national ICUs of other hospitals sharing their knowledge and adopting our findings - Set up the course modules for Master Classes for professionals for valorisation - Support the scientific and professional publication process for WPs - Branding of the flagship, public website, social media use - Network including at least 10 hospital partners, autonomously growing up to 15 - 20 by the end of the flagship		- Plan of action for the deployment of AIPT applications - Quarterly Master Classes for the hospital network and healthcare professionals - Publication plan for the Flagship - Website & newsbullets for public communications - Participation in health-tech exhibitions

Work Package 7 - Project management

The **objective** is to ensure consortium coordination, operational management, and technical vitality for the flagship, including administrative, contractual, financial, legal, technical, and ethical aspects. WP7 is connected to the broader flagship management infrastructure, ensuring compliance with reporting and other accountability and communication requirements and coordination with the independent Scientific Advisory Board for the flagship research (see 4.d.Governance).

Approach. Our governance structure safeguards project management quality and continuous communication with flagship advisors and Scientific Advisory Board.

Work Package 7		
Main Lead EMC	Partners TUD & EUR	Interconnected Work packages All
Main Aim	Ensure the financial and administrative and success of the flagship and connect partners and create harmony in collaborations.	
Tasks		Output
- Project administration and coordination; finance management and administration - Risk management; data, IP & knowledge management, ethical & safety management - Report to coordinating flagship board on milestones and deliveries, 2/year - Advisory board of independent (inter)national experts and maintain their input 2/y		- Research data management plan - Fiscal report - Publication report - Project report

Tables-3&4 outline the WP schedule/talent/staff requirements. WP leadership and PhD/postdoc supervision is equally distributed to benefit from each institution's track record. All PhDs/postdocs will work on cross-cutting topics to promote collaboration and generate a common language and understanding of AIPT co-creation, use and evaluation. Our work also depends on support staff, including students and professionals.

Table-3. Estimated staffing required.

Work Packages	Lead / Partner Institutions	PhD Research		Postdoc Research		Support / Student Involvement	Partners (March 2022)
		Flagship	Employed	Flagship	Employed		
WP1 – Community Building & Co-design	EUR EMC TUD	PhD1	PhD7	Postdoc1	Postdoc6	Student DreamTeam Bachelor/Master students Project Manager Nurses Patient and Family Representative	Treant Hospital ItéMedical Linksight <i>SAS</i> SyncVR Center for Safety in Healthcare Family Centered ICU Dutch Inspectorate Dutch Healthcare Institute <i>Dutch Healthcare Insurers</i>
WP2 – Data management, Infrastructure & Ethics	TUD EUR EMC	PhD2		Postdoc2		Bachelor students Data Trainers AI-Engineers IT Experts (EMC) Data Manager	ItéMedical Linksight <i>SAS</i>
WP3 – Human-centered AI Development	TUD EMC EUR	PhD3 PhD4	PhD8	Postdoc3	Postdoc7 Postdoc8	Bachelor/Master students AI-Engineers Technical Assistance Nurses	ItéMedical Linksight <i>SAS</i> Center for Safety in Healthcare
WP4 – Education & Professionalisation	EMC TUD EUR	PhD4		Postdoc4		Education Specialists Nurses	SyncVR <i>ErasmusX</i> <i>Erasmus MC Academy</i> <i>Leiden-Delft-Erasmus Center for Education</i>
WP5 – Formative Evaluation & AIPT Adoption	EUR TUD EMC	PhD5	PhD8	Postdoc5	Postdoc6	Patient & Family Representative Nurses Chief Clinical Advisor Chief Technical Advisor Ethical and Juridical Advisor (EJA) Bachelor/Master students	
WP6 – Dissemination & Exploitation	EMC EUR TUD	All	All	All	All	Project Manager Master Students	Treant Hospital <i>Other (Inter)National Hospitals</i>
WP7 – Project Management	EMC					Project Manager Master Students	
a) Text in bold face – PhD and Postdoc projects that are “must to have” and are secured funding by the Flagship partners and Flagship co-funding (PhDs for four years and Postdocs for 12 months to 5 years) and Partners that contributed in cash for the Flagship b) <i>Text in Italics</i> – partnership to be confirmed c) Existing PhDs and Postdocs have already been funded by Flagship relevant projects and collaborations							

Table-4. Timeframes for the estimated Flagship programme.

Work Packages	Year 1		Year 2		Year 3		Year 4		Year 5	
	0 – 6 m	6 – 12 m	12 – 18 m	18 – 24 m	24 – 30 m	30 – 36 m	36 – 42 m	42 – 48 m	48 – 54 m	54 – 60 m
WP1 – Community Building & Co-design	Community established	Co-design Phase 1			Co-design Phase 2		Co-design Phase 3			
	PhD7				PhD1					
	Postdoc6				Postdoc 1*					
WP2 – Data management, Infrastructure & Ethics	Data Management and Modelling Platform and Board			AIPT Model Development Phase 1		AIPT Model Development Phase 2		AIPT Model Development Phase 3		
	PhD2 – Use case 1*									
WP3 – Human-centered AI Development	Postdoc2									
		Design Requirements Received		AIPT development Phase 1	Phase 1 Testing	AIPT development Phase 2	Phase 3 Testing	AIPT development Phase 3	Phase 3 Testing	
	PhD3 – Use case 2*									
	PhD4 – Use case 3*									
	PhD8									
	Postdoc3									
	Postdoc7									
Postdoc8										
WP4 – Education & Professionalisation	Competence framework/s for AIPT use									
	Educational/training modules				Summer Schools					
	PhD4*									
WP5 – AI Adoption & Formative Evaluation	Postdoc4*									
	Ethnography of the Co-design processes									
	Ethnographic Study			Evaluation of AIPT adoption and Impact						
	PhD8				PhD5*					
	Postdoc5									
WP6 – Dissemination & Exploitation						AIPT Deployment in hospitals				
	Dissemination & Exploitation Plan				Master Classes					
WP7 – Project Management		SAB Meetings		SAB Meetings		SAB Meetings		SAB Meetings		SAB Meetings
	Program Management									
This table represents the ideal staffing for the flagship for 5 years.										
* With our current budget (March 2022), we will fund Postdoc 1 (WP1) and Postdoc4 (WP4, 1year); PhDs 2 (WP2), PhD3 and PhD4 (WP3), PhD4 also (WP4) for the use cases and PhD5 (WP5). The PhDs are co-supervised by the adjacent institutions.										
- <i>Italic text indicates PhDs and postdocs that have existing funding (i.e., funding excluded from Flagship co-finance).</i>										
- Remaining PhDs and postdocs require funding via NOW, EU-Horizon, etc. in the course of the flagship programme.										

3.d Alignment of the program with relevant national and international research agendas

At national level, our ambitions correspond with the Grand AI Research Challenges, part of the Artificial Intelligence Research Agenda for the Netherlands, especially Challenge 3 (AI Systems and Humans) and Challenge 4 (AI Systems and Society). The flagship fully embraces its vision of human-centered AI, developing AI approaches that augment rather than replace human intelligence and where humans and AI cooperate.

Our human-centric AI approach is compatible with the European ideal of using AI to protect fundamental human rights and public interests. Living labs, co-creation workshops and regular feedback through formative evaluation promote trustworthy AI technologies and help build professional skills and understanding (COM,2020). We align with the EU's principles and requirements for trustworthy AI, which emphasize human agency and oversight, technical robustness and safety, privacy and data governance, transparency, diversity, non-discrimination and fairness, environmental and societal well-being, and accountability (AI HLEG,2019).

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

Please note that the full list of applicants is on page 1.

Academic Leads

Prof.dr.D.Gommers, head of department and critical care specialist at the ICU for adult patients. During the COVID pandemic, he chaired the Dutch Society for Intensive Care. He has published some 200 peer-reviewed papers. He co-founded the ICU2.0 programme, focusing on Data- and AI-driven care at the AICU. He runs the Diederik Gommers ICU Foundation, €3M.

Dr.E.Özcan, director of Critical Alarms Lab at TU-Delft and Care Technology lead at EMC AICU focusing on humanising critical care technologies. She runs the Vital Soundscapes project with Philips Healthcare on alarm-nurse interactions and positive sounds for patient wellbeing and has generated close to €2M in funding for her research.

Dr.I.Wallenburg, trained nurse and associate professor of Care Sociology (EUR). She is academic lead of the Health programme in AiPact (€4M), a transdisciplinary research programme examining the societal impact of AI. She is co-founder of RN2Blend, a multi-centre study on nurse professionalisation in the Netherlands (€5M). She currently supervises ten PhD students and has supervised >50 Bachelor and Master students.

ErasmusMC

Prof.dr.M.de Hoog, paediatrician and director of PICU. He is an expert in (pharmacologic) prediction models and initiated the Dutch PICU/NICU lab, a collaboration with the Health Insurers Association. He is currently receiving funding for the Dutch Paediatric Formulary (Ministry of Health, €1.5M), Priority Medicines for Children (ZON-MW, €600k), and Dashboarding Parental Empowerment (Koers 23, EMC €55k).

Prof.dr.I.Reiss, head of the NICU and full professor of Neonatology. He has authored >350 peer-reviewed publications and currently supervises ten PhD students. In the past five years, he worked closely with private data and technology companies, receiving €2M in research funding.

Prof.dr.W.Niessen, expert in biomedical image analysis, AI, machine learning, imaging genetics and image-guided interventions. He has published widely in top international journals (>800 publications). He founded Quantib BV, responsible for bringing seven FDA-approved medical image analysis devices to market (now used in >100 hospitals worldwide) and awarded €69M in funding to establish National Health Data Infrastructure for Research & Innovation. He has supervised 60 PhD students and around 50 bachelor/master students.

Prof.dr.M.van Dijk, professor of Nurse Science and psychologist. She supervises PhD and nursing students. She is involved in various data-driven care projects and wider research initiatives regarding nursing care and nurse engagement and professionalisation. She is a member of the management board of the EMC Nurse Advisory Council.

Dr.M.van Genderen, critical care specialist and AI expert. He supervises a PhD student on AI development in critical care.

TUD

Prof.dr.A.Bozzon, professor of Human-Centred AI. He has published >150 papers in international peer-reviewed journals. He currently supervises 13 PhD students and has supervised >90 MSc and BSc students. He has acquired more than €5M in research funding from industry and local, national, and European funding bodies, with a total budget of more than €30M.

Prof.dr.M.Specht, expert in educational technologies, educational innovation and evaluation research. He is scientific director of the Leiden-Delft-Erasmus Centre for Education and Learning and heads the 4TU Centre for Engineering Education. He is also president of the European Association of Technology Enhanced Learning.

Prof.dr.J.Treur, full professor of AI and acknowledged expert on multidisciplinary human-like AI-modelling. He has published over 700 well-cited papers about cognitive, affective, and social modelling and AI systems making use of such models. He has also supervised >40 PhD students and edited three books on (adaptive) network-oriented AI-modelling and its application in various other disciplines.

Dr.J.Dauwels, associate professor in Circuits and Systems, Department of Microelectronics. He has supervised 100+ MSc students and 100+ BSc students over the past ten years, raised €10M in research funding, has 20+ patent applications worldwide, and has won several best paper awards at conferences and journals. His academic lab has spawned four start-ups across a range of industries, ranging from AI for healthcare to autonomous vehicles.

Dr.J.Yang, assistant professor at the Web Information Systems (WIS) group at TUDelft. He co-leads the Kappa research line on Crowd Computing & Human-Centred AI. He has attracted research funding from NWO on Trustworthy AI-based systems for sustainable growth (€95.7M in total, 2022-32), and for the TUD AI Lab (€1.06M).

EUR

Prof.dr.R.Bal, professor in Healthcare Governance at ESHPM. His studies include the use of data in governing healthcare organisations. Specialising in action-based research, he has established partnerships with the Healthcare Inspectorate and the Dutch Healthcare Institute to study the regulation of technologies and quality of care (€100K/year). Other recent grants related to action-research: Dancing with a virus (ZonMw, €350k) and RN2Blend (VWS, €5M).

Prof.dr.I.Leistikow, inspector at the Dutch Inspectorate for Healthcare&Youth and endowed professor of healthcare governance at ESHPM. His research revolves around patient safety, in particular work wellbeing and patient safety. He supervises PhD and master students.

Prof.dr.E.Stamhuis, chair for Law and Innovation at ESL. He researches the interaction between law, governance and new digital technologies, with a special focus on the public domain, healthcare and regulated markets. He is co-applicant for the AI-MAPS ELSA lab, awarded €2.1M. He is an invited expert in the Advisory Council of eHealth Junior, the Netherlands Cyber Security Lab, and Dutch expert group on AI-manipulated evidence in criminal cases.

Prof.dr.C.ter Hoeven, professor of Organizational Dynamics in the Digital Society at the Erasmus School of Social and Behavioral Sciences (ESSB). Her teaching and research focus on the human labour necessary to make AI work. She holds an ERC Consolidator Grant (2021-26) for The Ghostworker's Well-being: an Integrative Framework (€2M).

Dr.A.M.Weggelaar, registered nurse and associate professor of Change Management at ESHPM. Her action-research (in multiple forms) focuses on the most challenging transformations in healthcare: nurse craftsmanship, up-take of big data and digital integrated care. She led a work package in one of the largest EU-funded Big Data projects, BigMedilytics (€12M).

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Aim(1): Investigate how public and private collaboration can be organized and sustained to facilitate co-design of AIPT for critical care.

Prof.dr.Gommers, deHoog, and Reiss are highly experienced in collaborating with industrial partners and have attracted robust funding from private partners in recent years. Prof.dr.Bal and Dr.Wallenburg are experts in action-research and in collaborating with healthcare organizations and public regulatory agencies[32, 39, 40]. Prof.dr.Leistikow and Dr.Abrishami are closely allied with two key public healthcare regulators in this research and highly experienced in regulatory governance[41, 42]. Dr.Özcan is an expert in participatory design in AI[14].

Aim(2): Define the legal and ethical requirements for data collection and data use for AIPT creation and management, and how this can be organized in accordance with laws and regulations on (inter)national level. Prof.dr.Stamhuis and Dr.Dúran are experts in AI-related legal and ethical issues[43, 44], Prof.dr.Niessen, Prof.dr.Treur, Dr.Dauwels, Dr.Yang and Dr.Verma are experts in data and computational science, human-centred AI and/or predictive medicine[4, 28]. Dr.van Genderen is involved in data-infrastructure building and management at EMC. Prof.Niessen is an expert in open-source data and software resources. Together, they cover the complicated field of data ethics and AI regulation.

Aim(3): Alignment for AIPT development and use to reduce perceptual and cognitive load and interrupted workflows while increasing quality/effectiveness/outcomes of care and trust in AIPT among professionals/patients/family. Dr.Özcan is an expert in participatory design approaches in healthcare[45] and Prof.dr.Niessen, Prof.dr.Bozzon, Prof.dr.Treur and Dr.Verma are experts in co-design of AI and human-AI interactions[14, 27, 29, 46]. Dr.vanGenderen is critical care specialist and AI-medical researcher. Dr.Weheens and Dr.Weggelaar are experts in the adoption of complicated technologies in healthcare practice as well as action-oriented research[15, 47]. Dr.Weggelaar is also a registered nurse. Prof.dr.terHoeven and Dr.vanZoonen are experts in work & AI and communication science[48]. Prof.Niessen is experienced in bringing AIPT to the market.

Aim(4): Develop competencies, skills and educational methods to prepare current and future generations of healthcare professionals and students to enable human-centred AI development and use in critical care. Dr.Klijn is a critical-care specialist and specializes in professional education and training, Prof.dr.Specht and Dr.Limbu are experts in educational technologies and innovation (i.e., Virtual and Mixed Reality)[49]. Dr.Weggelaar is an expert in nurse professionalization[50].

Aim(5): Evaluate the impact of co-designed AIPT on critical care practice, the work of nurses/physicians and the experience of patients/family, as well as the conditions for valuable AI adoption in critical care. Dr. Wallenburg is an expert in formative evaluation and nursing and medical practice reform[32, 51], Dr.vanZoonen specialises in the impact of (AI) technology on work and work perception[30]. Prof.dr. van Dijk is an expert in nursing[52]. Dr.vanBommel is a critical care physician and expert in AI in medical practice. Dr.Sülz is an expert in performance metrics in relation to complex technologies[47]. Prof.Niessen and Prof.Bozzon, Dr.Yang are experts in human-centred AI[27].

Aim(6): Disseminate AIPT and knowledge for critical care to other hospitals and healthcare professionals, in accordance with ethical/professional/legal requirements, to facilitate wider learning and foster sustainability of future critical care provision. Prof.dr.Gommers, Prof.dr.Reiss, Dr.vanGenderen, Dr.Weggelaar and Drs.Mulder belong to a wider active community of hospitals working on AI.

4.c Involvement of partners

This flagship includes public partners, industrial partners as well as educational partners. They are all listed in Table-3. The ICUs work closely together with Dräger, ITéMedical, SAS, Chipsoft. ITéMedical has confirmed (bringing in 1.1M), others are under negotiation and willing to contribute. New partners have joined: SyncVR (particularly focusing on education) and Linksight (on global data usage for general purposes). Treant Zorggroep is a confirmed hospital partner that is ready to learn on AIPT adoption in critical care; we are currently discussing collaboration with other hospitals that have shown similar interest, also in the US (e.g., Northwell Group). Confirmed public partners are the Dutch Inspectorate, the Dutch Healthcare Institute and FCIC, the Family & Patient-Centered ICU. We have close contacts with the Dutch Health Insurers Association, Netherlands Healthcare Authority and other patient organizations. Also, we will work together with the ErasmusMC Academoy, ErasmusX and Medical Delta Centre for Education & Learning; three innovative education institutes we have warm connections with. We expect more (industrial) partners approaching us when the flagship is granted as we are offering an attractive ecosystem for the healthcare community.

4.d Governance, organization and collaboration strategy of the consortium

The flagship has **three Academic Leaders (AL)** from each participating institute who contribute to the flagship's core knowledge base and support the academic decision-making process (i.e., recruitment of PhD/postdoc candidates, publication plan and research agendas). A **Core Team of WP Leaders (CT)** ensures the success of each WP but also their interconnectedness and the dissemination and exploitation of knowledge domains. The CT also plays an active role in valorisation and further funding. The ALs and WP Leaders meet once a month. A **Project Manager** has an administrative role and mainly covers finance and the project timeline. The **Chief Technical Advisor (CTA)** is responsible for AI-Exploitation and Data Management and supports/facilitates data use, deployment, and possible implementation processes. The **Ethical and Juridical Advisor (EJA)** safeguards data usage from a legal and ethical perspective and ensure ethical treatment of participants in the co-design and validation studies. The CTA and EJA are part of WP2. The **Patient & Family Representative (PFR)** ensures the rights of the families in AIPT development and facilitates engagement with the Patient & Family Council for co-design and evaluation purposes. The PFR will be appointed in WP1. The **Chief Clinical Advisor (CCA)** links the use-cases to the clinical workflows and connect with nurses/teams for validation studies. The **Public Relations Advisor (PRA)** promotes a good public image and branding and is responsible for communication among various stakeholders (hospitals, industry, public actors, patient and family organisations, nurse associations). The PFR and CCA are part of WP1. The PRA also assists the project manager, both are listed in the budget. We will establish a **Scientific Advisory Board (SAB)** that is an independent observer monitoring progress towards our goals and missions that will meet the afore-mentioned leaders/experts/chiefs twice yearly.

We will also create **communities for PhDs/postdocs** who work on the flagship projects but also on related topics (i.e., PhDs associated with other funded and relevant projects) to foster more mutual academic and moral support.

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I Budget table (including the external funding table)

Flagship:	Human-Centered AI for Sustainable Critical Care: Co-designing AI-Powered Technologies in Neonatal, Pediatric and Adult Intensive Care		
Flagship acronym:	Critical Care		
Naam Penvoerder:	Erasmus MC		
Startdatum en looptijd van het project:	Startdatum 01-09-2022	Looftijd 5 jaar	
Totaal project budget:	€5.318.887		

[illegible]

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters

2a. Letter of intent Itémedical.pdf

2b. Letter of intent Linksight.pdf

2c. Letter of intent Treant.pdf

2d. Letter of intent SyncVR.pdf



Confidential

Erasmus MC
Prof. dr. Diederik Gommers
Doctor Molewaterplein 40
3015 GD Rotterdam

9 March 2022

LETTER OF INTENT
for the
SUSTAINABLE CRITICAL CARE PROJECT

Dear Professor Gommers,

I, Peter Hof, in my capacity of CEO at itēmedical BV hereby confirm that itēmedical BV intends to contribute to the sustainable critical care project, as applied for by the main applicant, Prof. dr. Diederik Gommers, head of ICU department Erasmus MC.

Itēmedical BV intends to provide an in-kind contribution for project management, medical software development & testing and MDR certification for new medical software product functionalities in the fields of medical alarms, dashboards, early warning scores, high resolution medical device data for research and external AI engines, representing a monetary value of € 1.184.264 and further detailed in the project proposal and budget form.

Yours sincerely,


itēmedical

Name: Peter Hof

Position: CEO

Date: 9 March 2022

Letter of Intent for a project partner



Erasmus MC
Prof. dr. Diederik Gommers
Doctor Molewaterplein 40
3015 GD Rotterdam

March 10th, 2022

LETTER OF INTENT
for the
SUSTAINABLE CRITICAL CARE PROJECT

Dear Professor Gommers,

I, Martine van de Gaar, in my capacity of CEO at LinkSight BV hereby confirm that LinkSight BV intends to contribute to the Sustainable Critical Care project, as applied for by the main applicant, Prof. dr. Diederik Gommers, head of ICU department Erasmus MC.

LinkSight is a TNO spin-off company that was founded in 2021, and focuses on bringing novel secure multiparty computation technology (MPC) to the healthcare market so that healthcare organisations can perform privacy-by-design data analytics over multiple, separate datasets without sharing sensitive patient information.

Being a recently founded startup company, LinkSight is in the first 2 years of the project not yet in the position to provide an in-cash contribution.

LinkSight intends to provide an in-kind contribution representing a monetary value of € 50.000,00 to € 100.000,00 (based on further discussion and detailing), which consists of:

- Access to the LinkSight MPC platform for privacy-by-design data analytics by the project partners (for this project specifically)
- Knowledge on MPC, privacy enhancing technologies, and decentralized data sharing architectures.

LinkSight BV
Veilingweg 61
3981 PB Bunnik



Erasmus MC
Prof. dr. Gommers
Doctor Molewaterplein 40
3015 GD Rotterdam

8 maart 2022

LETTER OF INTENT
for the
SUSTAINABLE CRITICAL CARE PROJECT

Dear prof. dr. Gommers,

I, [first name and family name], in my capacity of [position in the organisation (has to be a duly authorised person)] at Treant Zorggroep hereby confirm that Treant Zorggroep intends to contribute to the sustainable critical care project, as applied for by the main applicant, prof. dr. Gommers, head of ICU department Erasmus MC.

Treant Zorggroep intends to contribute € 15.000 in cash towards the project costs in accordance with the budget in the project proposal and budget form.

Yours sincerely,

Name:

R. Akkerman

Position:

Member of the board Treant group

Date:

15 MARCH 2022

Letter of intent for a project partner



Erasmus MC
Prof. dr. Diederik Gommers
Doctor Molewaterplein 40
3015 GD Rotterdam



15-03-2022

LETTER OF INTENT
for the
SUSTAINABLE CRITICAL CARE PROJECT

Dear Diederik Gommers,

I, Robbert Brouwer, in my capacity of co-founder and COO at SyncVR Medical hereby confirm that SyncVR Medical intends to contribute to the sustainable critical care project, as applied for by the main applicant, Prof. dr. Diederik Gommers, head of ICU department Erasmus MC.

SyncVR Medical B.V. intends to provide an in-kind contribution of consultancy hours and software licences, representing a monetary value of €50.000,- and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Robbert Brouwer

Position: co-founder & COO

Date: 15-03-2022

Letter of Intent SyncVR Medical B.V. - Padualaan 8 - 3584CH Utrecht



Erasmus MC
Prof. dr. Diederik Gommers
Doctor Molewaterplein 40
3015 GD Rotterdam



15-03-2022

LETTER OF INTENT
for the
SUSTAINABLE CRITICAL CARE PROJECT

Dear Diederik Gommers,

I, Robbert Brouwer, in my capacity of co-founder and COO at SyncVR Medical hereby confirm that SyncVR Medical intends to contribute to the sustainable critical care project, as applied for by the main applicant, Prof. dr. Diederik Gommers, head of ICU department Erasmus MC.

SyncVR Medical B.V. intends to provide an in-kind contribution of consultancy hours and software licences, representing a monetary value of €150.000,- and further detailed in the project proposal and budget form.

Yours sincerely,

A handwritten signature in black ink, appearing to read "Robbert Brouwer", written over a horizontal line.

Name: Robbert Brouwer

Position: co-founder & COO

Date: 15-03-2022

Letter of Intent SyncVR Medical B.V. - Padualaan 8 - 3584CH Utrecht

III Letter of support from department head of the main lead

On headed paper of the department or institution.

Application form Flagship program round 2021-2022


 Health &
Technology
By Convergence

Dr. Rik Wehrens	EUR	Healthcare Institute Healthcare Governance	wehrens@eshpm.eur.nl	Tenured	n.a.	Applicant
Dr. Anne Marie Weggelaar	EUR	Health Services Management & Organisation Erasmus School of Law	weggelaar@eshpm.eur.nl	Tenured	n.a.	Applicant
Prof.dr. Evert Stamhuis	EUR	Organizational Dynamics in the Digital Society	stamhuis@law.eur.nl	Tenured	n.a.	Applicant
Prof.dr. Claartje ter Hoeven	EUR	Organizational Dynamics in the Digital Society	terhoeven@essb.eur.nl	Tenured	n.a.	Applicant
Dr. Ward Zoonen	EUR	Organizational Dynamics in the Digital Society	w.vanzoonen@essb.eur.nl	Tenured	n.a.	Applicant
Dr. Sandra Sülz	EUR	Health Services Management & Organisation	sulz@eshpm.eur.nl	Tenured	n.a.	Applicant
Dr. Marjolein Timmers	EUR	Law & Healthcare	timmers@eshpm.eur.nl	Tenured	n.a.	Applicant
Dr. Ernst Hulst	EUR	Law & Healthcare	hulst@eshpm.eur.nl	Tenured	n.a.	Applicant
Dr. André den Exter	EUR	Erasmus School of Law	denexter@law.eur.nl	Tenured	n.a.	Applicant

Corresponding Lead: Prof. Dr. Diederik Gommers

Planned start date of seed funding: 1-9-2022

Primary Convergence theme: *Human Centered Technology and AI for Health*

Secondary Convergence theme (if applicable): *Technology Supported Transitions in Health(care)*

Tertiary Convergence theme (if applicable): *Improving Health Journeys*

Quaternary Convergence theme (if applicable): Choose an item.

Keywords: critical care, nurses, workforce sustainability, human-centered AI, data ethics, participatory design, healthcare governance

voet afkloord,

1.b Summary (max 300 words)

One of the most pressing problems in contemporary healthcare systems is the growing shortage of healthcare staff, particularly specialized nurses. Beyond nurse education and professional leadership, technological solutions are urgently needed to enhance the sustainability of critical care. In this flagship, which involves neonatal, paediatric and adult ICUs, we shift from mechanically assisted to AI-powered critical care and empower ICU nurses and physicians by

J.G. Boonstra